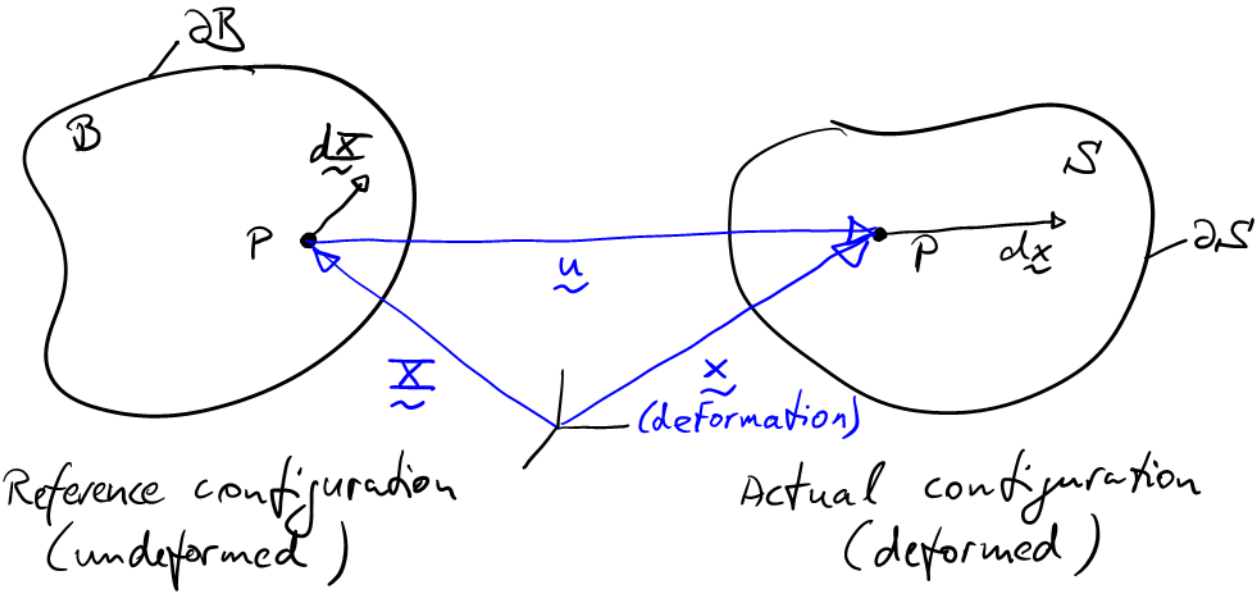
Introduction to Continuum Mechanics

Kinematics

- Continuum: set of material points, which continuously fill a physical domain (body)



Displacement $\underset{\sim}{u} = \underset{\sim}{x} - \underset{\sim}{X}$

- Deformation gradient:

$$d\underset{\sim}{x} = \frac{\partial \underset{\sim}{x}}{\partial \underset{\sim}{X}}\, d\underset{\sim}{X}$$

$$=: \underset{\sim}{F} = \mathrm{Grad}\, \underset{\sim}{x}$$

cap. "G" for derivative w.r.t. $\underset{\sim}{X}$

- Multiplicative split:

$$\underset{\sim}{F} = \underset{\sim}{R}\,\underset{\sim}{U}$$

$\underset{\sim}{U}$: stretch tensor

$\underset{\sim}{R}$: rotation tensor (orthogonal, $\underset{\sim}{R}^T = \underset{\sim}{R}^{-1}$)

→ contains rotations

- Right Cauchy-Green deformation tensor:

$$\underline{\underline{C}} := \underline{\underline{F}}^T \underline{\underline{F}}$$

$$= (\underline{\underline{R}}\,\underline{\underline{U}})^T \underline{\underline{R}}\,\underline{\underline{U}} = \underline{\underline{U}}^T \underline{\underline{R}}^T \underline{\underline{R}}\,\underline{\underline{U}} = \underline{\underline{U}}^T \underline{\underline{R}}^{-1} \underline{\underline{R}}\,\underline{\underline{U}} = \underline{\underline{U}}^T \underline{\underline{U}}$$

$\rightsquigarrow$ does not contain rotations ✓

Reference configuration

$\rightsquigarrow$ zero deformation

$\rightsquigarrow$ $d\underset{\sim}{x} = \underline{\underline{1}}\, d\underset{\sim}{X}$

$\hookrightarrow \underline{\underline{F}} = \underline{\underline{1}}$

$\rightsquigarrow \underline{\underline{C}} = \underline{\underline{F}}^T \underline{\underline{F}} = \underline{\underline{1}}^T \underline{\underline{1}} = \underline{\underline{1}} \neq \underline{\underline{0}}$ (↯)

- Green-Lagrange strain tensor

$$\underline{\underline{E}} := \frac{1}{2}\left(\underline{\underline{C}} - \underline{\underline{1}}\right)$$

Ref. conf.: $\underline{\underline{C}} = \underline{\underline{1}} \rightsquigarrow \underline{\underline{E}} = \frac{1}{2}\left(\underline{\underline{1}} - \underline{\underline{1}}\right) = \underline{\underline{0}}$ ✓

- Representation in $\underset{\sim}{u}$:

$$\underset{\sim}{x} = \underset{\sim}{X} + \underset{\sim}{u} \rightsquigarrow \underline{\underline{F}} = \frac{\partial \underset{\sim}{x}}{\partial \underset{\sim}{X}} = \frac{\partial(\underset{\sim}{X} + \underset{\sim}{u})}{\partial \underset{\sim}{X}} = \underline{\underline{1}} + \mathrm{Grad}\,\underset{\sim}{u}$$

$$\rightsquigarrow \underline{\underline{E}} = \frac{1}{2}\left(\underline{\underline{F}}^T \underline{\underline{F}} - \underline{\underline{1}}\right)$$

$$= \frac{1}{2}\left(\left(\underline{\underline{1}} + \mathrm{Grad}\,\underset{\sim}{u}\right)^T \left(\underline{\underline{1}} + \mathrm{Grad}\,\underset{\sim}{u}\right) - \underline{\underline{1}}\right)$$

$$= \frac{1}{2}\left(\underline{\underline{1}}^T \underline{\underline{1}} + \underline{\underline{1}}^T \mathrm{Grad}\,\underset{\sim}{u} + \mathrm{Grad}^T \underset{\sim}{u}\,\underline{\underline{1}} + \mathrm{Grad}^T \underset{\sim}{u}\,\mathrm{Grad}\,\underset{\sim}{u} - \underline{\underline{1}}\right)$$

negligible for small $\underset{\sim}{u}$

→ Geometrically linearized theory

only valid for small displacements !!!

considered in this class !!!

$$\rightsquigarrow \underline{\underline{\varepsilon}} := \frac{1}{2}\left(\mathrm{Grad}\,\underset{\sim}{u} + \mathrm{Grad}^{T}\,\underset{\sim}{u}\right)$$

$$= \mathrm{sym}\left[\mathrm{Grad}\,\underset{\sim}{u}\right] \rightsquigarrow \underline{\underline{\varepsilon}} = \underline{\underline{\varepsilon}}^{T}$$

→ $\mathcal{S} \approx \mathcal{B}$, $\underset{\sim}{u}$ small, but $\underset{\sim}{u} \neq \underset{\sim}{0}$!

↳ no differentiation between $\underset{\sim}{x}$ and $\underset{\sim}{X}$

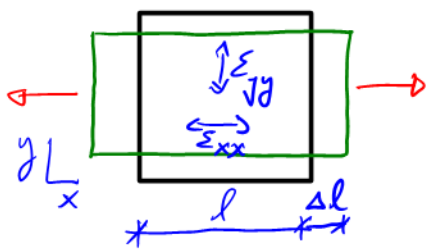
↳ $\underset{\sim}{X} \Leftarrow \underset{\sim}{x}$

$$\underline{\underline{\varepsilon}} = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13} \\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{23} \\ \varepsilon_{31} & \varepsilon_{32} & \varepsilon_{33} \end{bmatrix} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz} \end{bmatrix}$$

with the individual coefficients

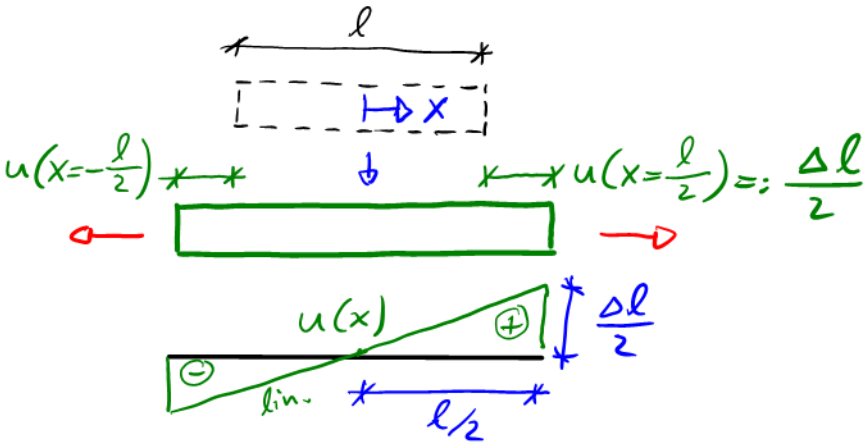
- $\varepsilon_{xx} =: \varepsilon_x = \frac{1}{2}\left(\frac{\partial u_x}{\partial x} + \frac{\partial u_x}{\partial x}\right) = \frac{\partial u_x}{\partial x} =: \underbrace{u_{x,x}}$ (simplified notation for derivative)

  ↳ normal strains



$\rightsquigarrow \varepsilon_{yy}, \varepsilon_{zz}$ analogous

Expl 1D:

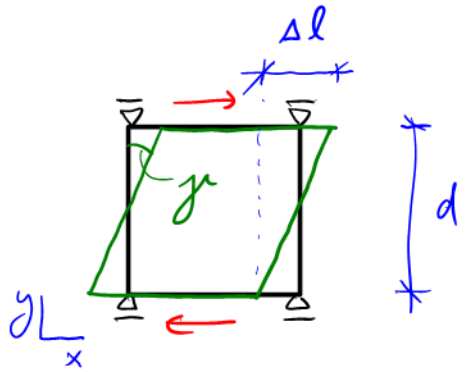


→ strains $\varepsilon(x) = \frac{\partial u(x)}{\partial x} = \text{const.}$

$$= \frac{u(x=\frac{l}{2})}{\frac{l}{2}} = \frac{\Delta l/2}{l/2} = \frac{\Delta l}{l}$$

- $\varepsilon_{xy} = \frac{1}{2}\left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right) = \frac{1}{2}\left(u_{x,y} + u_{y,x}\right) =: \frac{1}{2}\gamma_{xy}$

  ε_{xy}: shear strains; γ_{xy}: slip



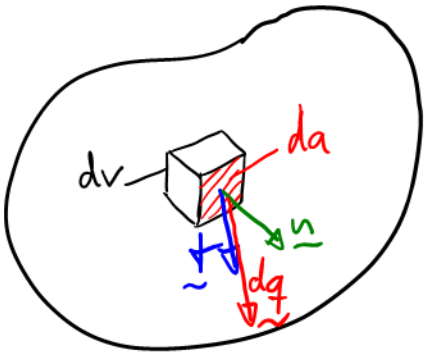
$\gamma_{xy} = \frac{\Delta l}{d}$

$\triangleq$ glide angle

$\underline{\varepsilon} = \underline{\varepsilon}^T \rightsquigarrow \varepsilon_{yx} = \varepsilon_{xy}$

ε_{yz}, ε_{zy}, ε_{xz}, ε_{zx} analogous

Stress state



$\underset{\sim}{n}$ ≙ normal vector

$d\underset{\sim}{q}$ ≙ resultant force in da

$\underset{\sim}{t}$ ≙ stress vector := $d\underset{\sim}{q}/da$

≙ average distributed force over da

⇝ dependent from $\underset{\sim}{n}$ ↯

Cauchy Theorem:

$$\underset{\sim}{t} = \underline{\sigma}\,\underset{\sim}{n}$$

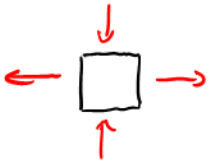
└ stress tensor ⇝ independent from $\underset{\sim}{n}$ ✓

⇝ tensor of 2nd order

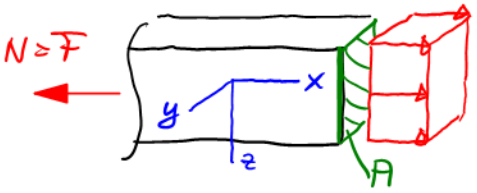
$$\underline{\sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}$$

Normal stresses

σ_{xx}, σ_{yy}, σ_{zz} ⇝



Expl. 1D:



$$\sigma(y,z) = \frac{N}{A} = \text{const.}$$

$\sigma > 0$ ⇝ tension

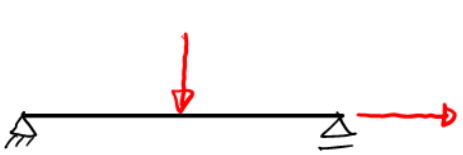
$\sigma < 0$ ⇝ compression

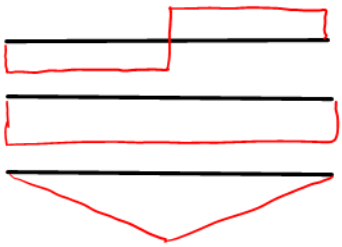
Shear stresses

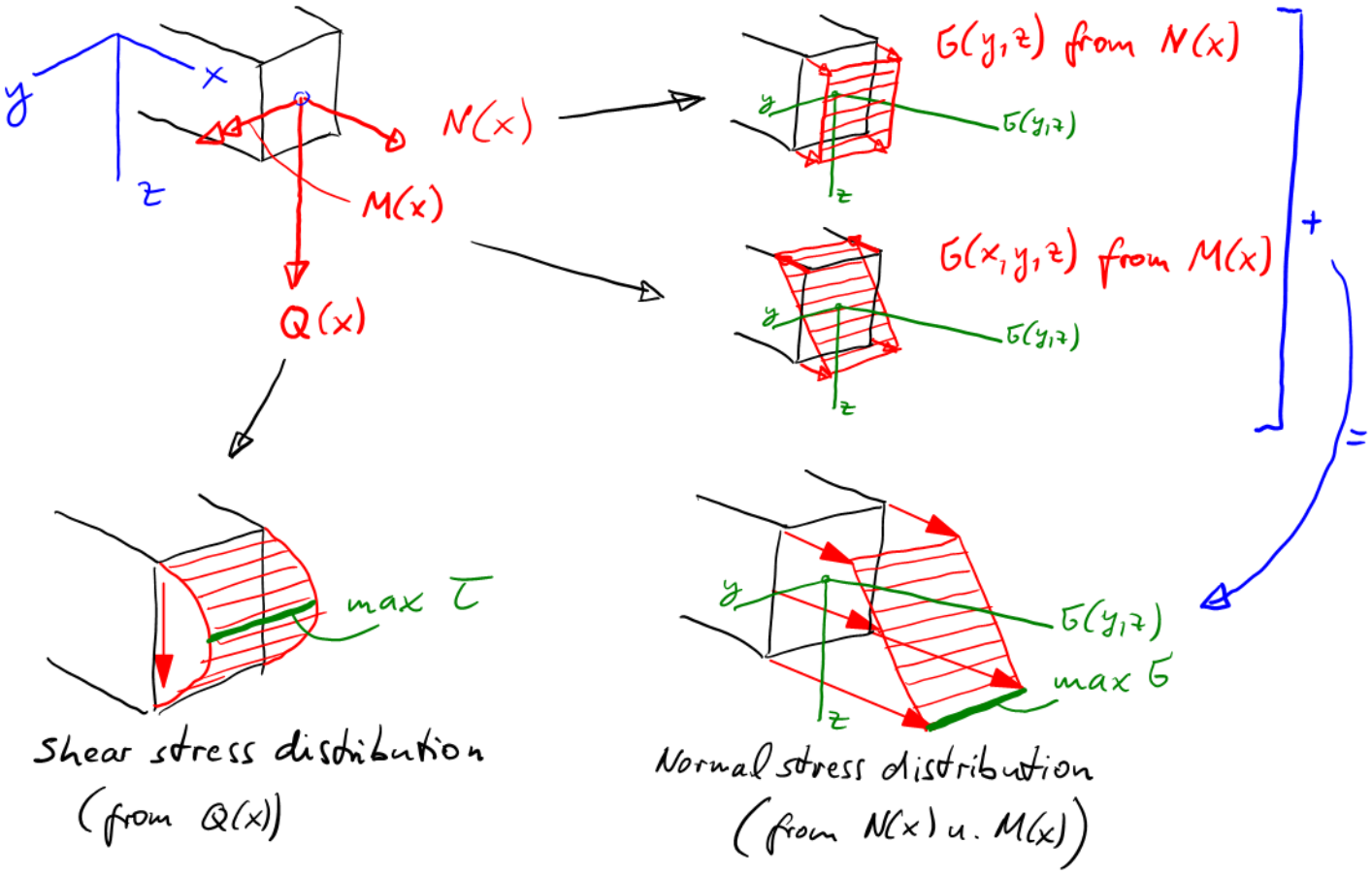
$\sigma_{xy}, \sigma_{yx}, \sigma_{xz}, \sigma_{zx}, \sigma_{yz}, \sigma_{zy}$ ⇝



Application to Engineering Mechanics







Shear stress distribution
(from Q(x))

Normal stress distribution
(from N(x) u. M(x))